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Introduction

The purpose of this lab was to study how RF signals behave in free space and how different materials affect the signal strength. We tested three antenna systems at 2.4 GHz and 935 MHz. We measured the received power in free space and then placed different objects (a person, glass, and wood) between the antennas to see how much signal was lost. We also tested how polarization affects the signal by rotating the horn antenna.

- **Setup**
- Signal generator - transmit antenna
- Receive antenna - spectrum analyzer
- Distance between antennas = 3 ft (0.914 m)
- Antennas used:
- 2.4 GHz Yagi-Uda
- X-band horn pair
- 935 MHz dipole

Measurements were taken by:

- Keeping distance fixed
- Adding obstructions
- Rotating the horn antenna (for polarization test)

Results:

Fre- Space and Obstruction measurments

Antenna	Freq	TX(dbm)	Free space	Person	Glass	Wood	FSPL (dB)
Yagi 2.4 GHz	2.4 GHz	-17.4	-85.70	-93.50	-88.60	-87.40	39.27
X-band horn Pair	2.4 GHz	+6.86	-54.75	-87.50	-61.40	-61.92	39.27
935MHz Dipole	935 MHz	-21.44	-55.70	-71.54	-58.70	-58.80	31.08

The Horn antenna had the strongest received signal in free space, while the yagi had the weakest.

Material Obstruction Loss:

Antenna	Person Loss	Glass Loss	Wood Loss
Yagi	-7.80 dB	-2.90 dB	-1.70
Horn	-32.75 dB	-6.65 dB	-7.17
Dipole	-15.84 dB	-3.00	-3.10

The person caused the biggest signal loss in all cases.

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Horn Antenna Polarization Sweep:

Rotation angle	Receiving power	From 0 degree (dB)
0 (aligned)	-54.75	0.00
15	-57.11	-2.36
45	-61.32	-6.57
75	-71.20	-16.45
90 (null)	-80.10	-25.35

As the antenna rotated, the signal dropped. The biggest drop happens at 90 degrees when the antennas are perpendicular.

Friis Path Loss Comparison:

Antenna	TX (dBm)	G_tx (dBi)	G_rx (dBi)	FSPL (dB)	Cable Loss (dB)	Pred. Rx (dBm)	Meas. Rx (dBm)	Δ (dB)
2.4 GHz Yagi-Uda	-17.4	10	6	39.27	1.5	-42.17	-85.70	-43.53
X-band Horn Pair	+6.86	15	15	39.27	1.0	-3.41	-54.75	-51.34
935 MHz Dipole	-21.44	2.15	6	31.08	1.5	-45.87	-55.70	-9.83

Discussion

Free-Space Path Loss:

At 3 feet (0.914 m), the path loss was higher at 2.4 GHz (about 39 dB) compared to 935 MHz (about 31 dB). This makes sense since higher frequencies lose more power over distance. The measured values were all lower than those predicted by the Friis equation. This is expected because the real setup has losses from cables, connectors, and small errors in antenna gain. Additionally, since the distance is relatively short, some measures may not yet be fully in the far field.

Obstruction Effects:

Out of all the materials, the person caused the biggest drop in signal. This was especially noticeable with the horn antenna at 2.4 GHz, where the loss was very large. This happens because the human body absorbs RF signals. At 935 MHz, the signal did not drop as much, which shows that lower frequencies can pass through objects better. Glass and wood only caused small changes in signal (just a few dB), so they do not block the signal as much compared to a person.

Horn Polarization:

When the horn antenna was rotated, the signal got weaker as the angle increased. The strongest signal was when both antennas were aligned. At 90°, the signal dropped a lot, meaning the antennas were no longer aligned properly. This is similar to what was seen in the previous lab. The drop was a little more than expected at some angles, probably because the horn antenna has a narrow beam.

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Frequency and Penetration:

The 935 MHz antenna showed less signal loss through the person compared to the 2.4 GHz antennas. This confirms that lower frequency signals can go through objects better. The horn antenna had a much bigger drop compared to the Yagi at the same frequency. This is likely because the horn is more directional, so blocking it has a bigger effect. For glass and wood, the results were more consistent across all antennas since those materials affect the signal in a similar way.

Conclusion:

In this lab, we tested how RF signals behave using different antennas and materials. The results showed that signal strength decreases due to distance and depends on frequency. The 2.4 GHz systems had more path loss compared to the 935 MHz system. The obstruction tests showed that a person causes the most signal loss, while glass and wood only cause small attenuation. This shows that the human body absorbs RF signals more than common materials. The polarization test showed that alignment between antennas is very important. When the antennas were rotated, the signal dropped a lot, especially at 90°, where the antennas are perpendicular.

Overall, the lab followed expected RF behavior. Even though the measured values did not exactly match the theoretical predictions, the trends were still correct. This lab showed how real-world factors like obstruction, frequency, and alignment affect signal strength.